HW 3 due: 2/2/24

1) $6 \cdot 7^n - 2 \cdot 3^n$ div by 4 for all $n \geq 1$

Base Step:

$n=1$: $6 \cdot 7^1 - 2 \cdot 3^1 = 42 - 6 = 36$ $36/4 = 9$ ✓

Inductive Step:

Suppose $6 \cdot 7^n - 2 \cdot 3^n$ is divisible by 4.

$6 \cdot 7^{n+1} - 2 \cdot 3^{n+1}$

$= 6 \cdot 7^n \cdot 7 - 2 \cdot 3^n \cdot 3$

$= (4+3)(6 \cdot 7^n) - 2 \cdot 3 \cdot 3^n$

$= 4 \cdot 6 \cdot 7^n + 3 \cdot 6 \cdot 7^n - 3 \cdot 2 \cdot 3^n$

$= 4 \cdot 6 \cdot 7^n + 3(6 \cdot 7^n - 2 \cdot 3^n)$ ($6 \cdot 7^n - 2 \cdot 3^n$: div by 4)

$4 \cdot 6 \cdot 7^n$ is divisible by 4 and $3(6 \cdot 7^n - 2 \cdot 3^n)$ is (by our assumption), so it divides their sum as well. By the inductive assumption, $6 \cdot 7^n - 2 \cdot 3^n$ is divisible by 4 for all $n \geq 1$.

2) $2n+1 \leq 2^n$, $n = 3, 4, \ldots$

Base $2(3)+1 \stackrel{?}{\leq} 2^3$

$7 \leq 8$ ✓

Induc Suppose $2n+1 \leq 2^n$.

For $n+1$:

$2(n+1)+1 \leq 2^{n+1}$

$2n+3 \leq 2^{n+1}$

Using our assumption: $2n+1 \leq 2^n$ (mult. each side by 2)

$4n+2 \leq 2^{n+1}$

If $2n+3 \leq 4n+2$, $2n+3 \leq 2^{n+1}$

$1 \leq 2n$

$\frac{1}{2} \leq n$ This is always true, for $n \geq 3$ so since

$2n+3 \leq 4n+2 \leq 2^{n+1}$

$2n+1 \leq n2^n$

3) B1) $\sum_{i=1}^{3} w_i = w_1 + w_2 + w_3 = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} = \frac{9}{12} = \boxed{\frac{3}{4}}$

$w_1 = \frac{1}{1} - \frac{1}{1+1} = \frac{1}{2}$

$w_2 = \frac{1}{2} - \frac{1}{2+1} = \frac{1}{6}$

$w_3 = \frac{1}{3} - \frac{1}{3+1} = \frac{1}{12}$

B1) $w_1 = \frac{1}{2}$

$w_2 = \frac{1}{6}$

$w_3 = \frac{1}{12}$

$w_4 = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$

$w_5 = \frac{1}{5} - \frac{1}{6} = \frac{1}{30}$

$w_6 = \frac{1}{6} - \frac{1}{7} = \frac{1}{42}$

$w_7 = \frac{1}{56}$

$w_8 = \frac{1}{72}$

$w_9 = \frac{1}{90}$

$w_{10} = \frac{1}{10} - \frac{1}{11} = \frac{1}{110}$

$\sum_{i=1}^{10} w_i = \boxed{\frac{10}{11}}$

sum of

$\frac{1}{1} - \frac{1}{1+1} +$

$C_n = \sum_{i=1}^{n} w_i = \frac{1}{n} - \frac{1}{n+1}$

$w_1 + \ldots w_n$

w_1

$\frac{1}{1} - \frac{1}{n+1}$

B2) $C_n = \frac{n}{n+1}$

For $n=1$: $\sum_{i=1}^{1} w_i = \frac{1}{1} - \frac{1}{1+1} = \boxed{\frac{1}{2}}$ ✓ $= \frac{1}{1+1}$

suppose $C_n = \frac{n}{n+1}$

$C_{n+1} = \sum_{i=1}^{n+1} w_i = w_1 + \ldots + w_{n+1}$

$C_{n+1} = C_n + w_{n+1}$

$C_{n+1} = \frac{n}{n+1} + \frac{1}{n+1} - \frac{1}{(n+1)+1}$

$C_{n+1} = \frac{n+1}{n+1} - \frac{1}{n+2}$

$C_{n+1} = 1 - \frac{1}{n+2}$

$C_{n+1} = \frac{n+2-1}{n+2}$

$C_{n+1} = \frac{n+1}{n+2}$

This is of the form $\frac{k}{k+1}$ (but $k = n+1$), proving our formula true.

$f(\alpha) = \alpha\alpha^R$

4) WTS one-to-one:
Must show that if $f(\alpha) = f(\beta)$, then $\alpha = \beta$. Suppose $f(\alpha) = f(\beta)$. Using the definition of f, this means $\alpha\alpha^R = \beta\beta^R$
Let $\alpha \cdot \alpha^R = K$ and $\beta\beta^R = \lambda$. If $\alpha\alpha^R = \beta\beta^R$, $K = \lambda$.
Thus, $K_0 = \lambda_0$
$K_{n-1} = \lambda_{n-1}$ and so on for length K and λ
This would mean $\alpha_n = \beta_n$ for their lengths, and $\alpha = \beta$.
Onto: NO, f is not onto. To be onto, every element in the codomain (set of palindromes over x must be hit. 'abbbba' is a palindrome and reads the same forwards and back, so it is in the codomain. However, there exists no α where $f(\alpha) = \alpha\alpha^R = abbbba$. The output of f is always an even length, as the length is $2 \cdot |\alpha|$ and 2 times anything is always even. |abbbba| is odd, and thus it'll never be hit by f, despite being in the codomain. Thus, f is not onto.

5) $(x, y) \in R$ if 3 divides $x - y$
✓ Reflexive: $3 \mid (x-x) = 3 \mid (1-1)$ → 0
Always 3 dividing 0, 3 does divide 0 since $0 = 3 \cdot k$.
$\forall x \in R$, xRx so it is reflexive.
✓ Symmetry: If xRy, then 3 divides $x-y$. 3 also divides $y-x$, as this would be $= -(x-y)$, and if an integer divides an integer, it also divides that integer negative.
$\forall x, y \in R$ if xRy, then yRx, so this is symmetric.
✓ Transitive: If xRy, 3 divides $x-y$. If yRz, then 3 divides $y-z$. This means 3 divides their sum → $(x-y)+(y-z) = x-z$. Thus, xRz.
$\forall x, y, z \in R$, if xRy and yRz, then xRz, so this relation is transitive.
X This relation is not partial order since it is not antisymmetric, even though it is reflexive and transitive

X (a) Reflexive: xRx? $x^2+x^2=1$

$2x^2=1$

$x^2=\frac{1}{2}$

This is never true for any integers, so the relation is not reflexive.

✓ Symmetric: If xRy, $x^2+y^2=1$. You can simply switch x and y → $y^2+x^2=1$. This is the same equation and thus true. $\forall x, y \in \mathbb{Z}$, xRy implies yRx, so the relation is symmetric.

X Transitive:

xRy $x^2+y^2=1$ } xRz ?

yRz $y^2+z^2=1$ } $x^2+z^2=1$

Add the 2 eqs: xRz $x^2+2y^2+z^2=2$

$x^2+z^2+2y^2=2$

Assume for sake of contradiction xRz. You can substitute x^2+z^2 as 1.

$1+2y^2=2$

$2y^2=1$

$y^2=\frac{1}{2} \Rightarrow y=\pm\sqrt{\frac{1}{2}}$ This is never true for any $y \in \mathbb{Z}$.

Thus, for $x, y, z \in \mathbb{Z}$, if xRy and yRz, that does not mean xRz. Relation is not transitive.

X Antisymmetric: To be antisymmetric, if xRy and yRx, then $x=y$.

xRy $x^2+y^2=1$

yRx $y^2+x^2=1$

if $x=y$, $x^2+x^2=1$

$2x^2=1$

$x^2=\frac{1}{2}$

$x=\pm\sqrt{\frac{1}{2}}$

This is not true for any x in $\mathbb{Z}$, so if xRy and yRx, then $x=y$, so relation is not antisymmetric.

Counterexample: $x=0, y=1$

xRy $0^2+1^2=1$ ✓

yRx $1^2+0^2=1$ ✓

However $0 \neq 1$, or $x \neq y$.

Therefore, this relation is not antisymmetric.

7) {1,2,3,4} ~~{(x,y) ∈ R | x² + y² = 1}~~ for every $x \in X$, xRx so reflexive.

{(1,1), (2,2), (3,3), (4,4), (1,2), (2,3)}

Not symmetric since (1,2) in set but not (2,1).

Not transitive since (1,2) and (2,3) in set but not (1,3).

8) This statement claims ~~=~~ that if the relation is symmetric ($xRy \Rightarrow yRx$), then xRx if it is transitive (if xRy and yRx) $\Rightarrow xRx$.

What this does not consider is that even if it's symmetric and transitive, this only applies to elements in the relation. If certain elements of X are missing from the relation, $(x,x) \notin R$. In order for the relation to be reflexive, $\forall x \in X$, xRx. xRx must be true for all elements in the set which is not guaranteed by its symmetric or transitive properties.

ex. $X = \{1,2,3\}$ $R = \{(1,1),(2,2),(1,2),(2,1)\}$ transitive ✓ symm ✓

(3,3) is not in the relation, so it is not reflexive.

5) Antisymmetry: Let $x=4$ and $y=1$.

xRy since 3 divides 4-1

yRx since 3 divides 1-4

However, $x \neq y$ since $4 \neq 1$. Therefore, the relation is not antisymmetric.